

Output – Experimental Results

Console-style summary of the run of student_search.py (all correctness assertions passed) followed by the generated plots.

All correctness tests PASSED (edge cases, cross-validation, per-dataset checks)

n	cmp	lin	bin	hash	us	lin	bin	hash	sort	ms	hash	ms
1,000		742	9.6	1.06		18.4	1.06	0.21		0.06		0.1
10,000		7,543	12.8	1.17		201.2	1.49	0.21		1.37		1.4
100,000		78,062	16.3	1.08		2,203.9	1.92	0.21		19.71		44.1
1,000,000		740,279	19.5	1.03		22,553.3	2.44	0.21		301.18		1216.0

Task 12 (n=1,000,000; 50,000 requests): binary 0.310 s | hash 0.065 s | linear (est.) 1222 s
Daily rebuild: re-sort 0.46 s | re-hash 1.98 s

Figure 1: Average number of comparisons vs n

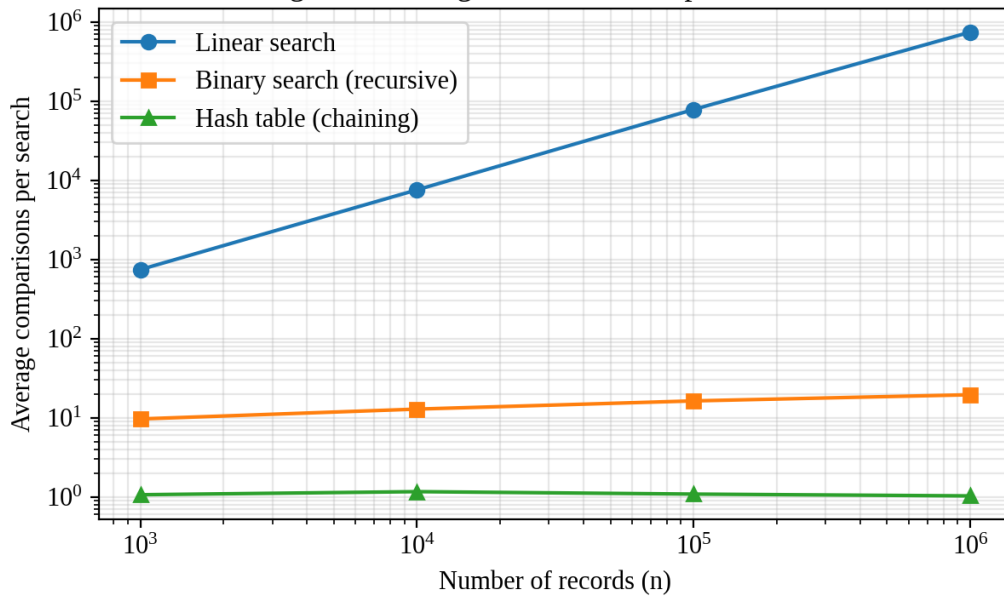


Figure 2: Average execution time per search vs n

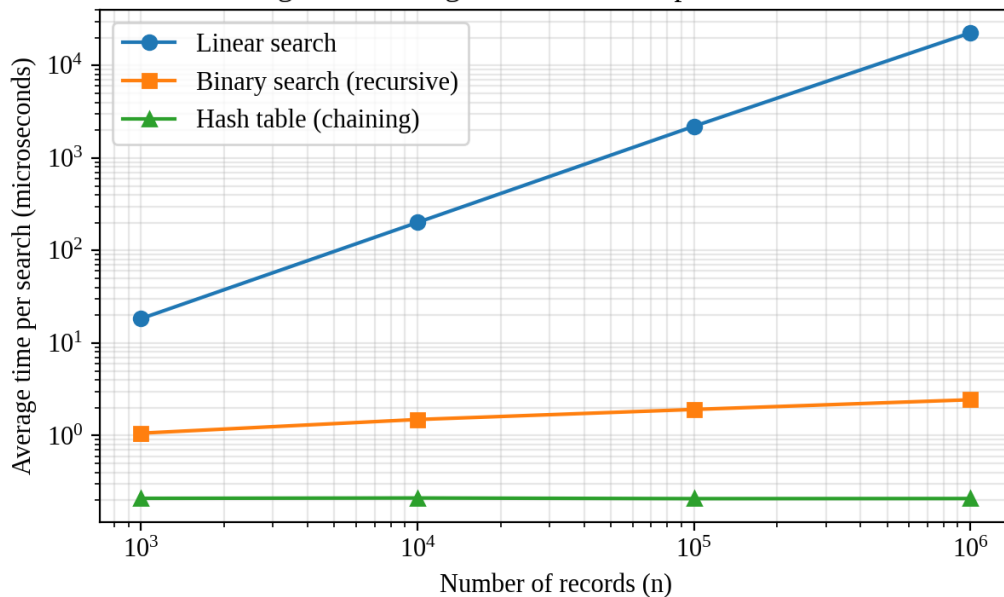


Figure 3: Preprocessing time vs n

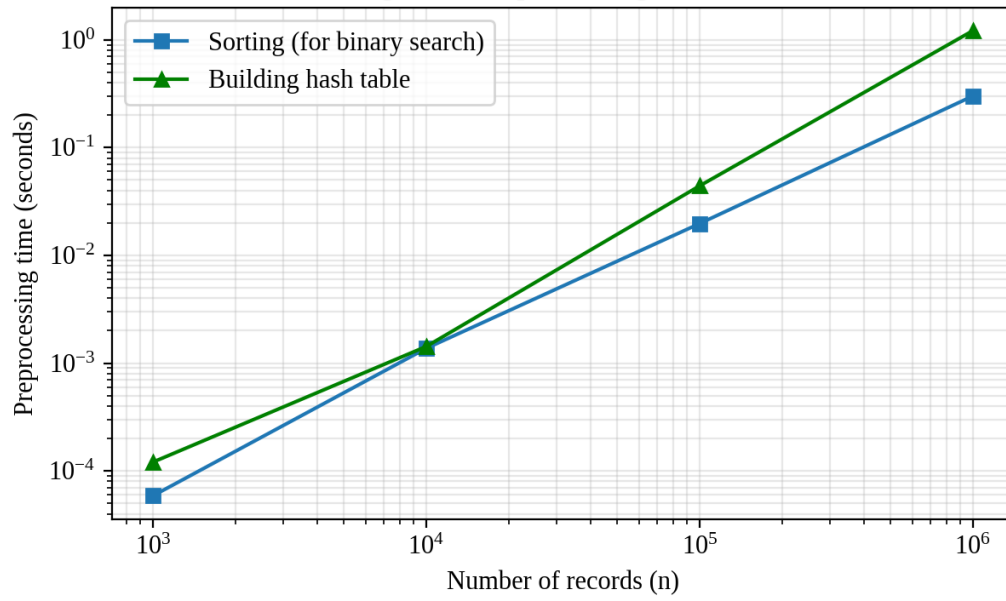


Figure 4: Total cost vs number of searches (n = 1,000,000)

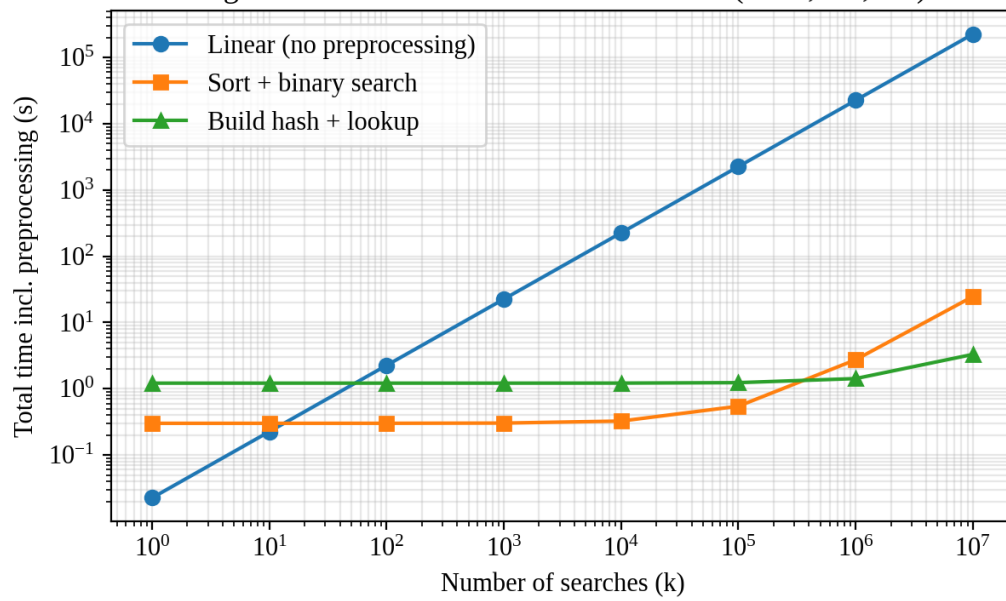


Figure 5: One hour of traffic (50,000 searches, $n = 1,000,000$)

